Automated Market Making: the case of Pegged Assets

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joint work with

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1. Introduction

Automated Market Makers (AMM)

- AMMs are decentralized protocols in which
  - 2 or more assets are maintained as reserves
  - Liquidity Providers (LPs) deposits liquidity in reserves assets
  - Liquidity Takers (LTs) can swap reserve assets
  - No counterparty risk
- Pricing:
  - Uniswap-like : simple pricing rules (CFMM + range LP)
  - CoW Swap-like : periodic batch auctions
  - **Swaap-like** : RFQ AMM with hybrid architecture
- Pegged assets:
  - Stablecoins: DeFi's reference assets - all pegged to the same fiat currency
  - LST: asset pegged to an underlying + yield

Our contribution

1. Build a novel type of price process to model the dynamics of pegged assets
   $\implies$ multi-level Ornstein-Uhlenbeck with filtering and calibration methods
2. Derive optimal quotes for market making assets following multi-level Ornstein-Uhlenbeck
3. Test the performance of such a trading strategy both on simulated and real data

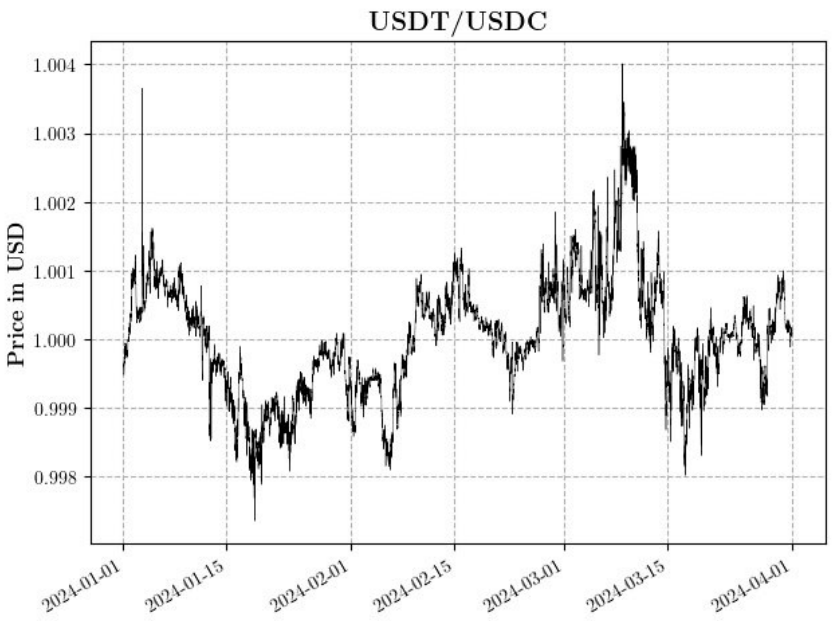
2. Price dynamics for pegged assets

Standard Ornstein-Uhlenbeck

- The most used mean-reverting process for assets dynamics is the Ornstein-Uhlenbeck process.

$$dS_t = -\kappa(S_t - \bar{S})\,dt + \sigma\,dW_t^S,$$

With S_t being the price of an asset at time t, $\bar{S}$ the long term target, dW_t^S standard brownian increments, and $\kappa > 0$ and $\sigma > 0$ parameters.



Multi-level Ornstein-Uhlenbeck

- We now consider that the long-term mean itself follows a Ornstein-Uhlenbeck process, that is we have

$$\begin{cases} dS_t = -\kappa \left(S_t - U_t\right)\, dt + \sigma\, dW_t^S, \\ dU_t = -\eta \left(U_t - \bar{U}\right)\, dt + \nu\, dW_t^U, \end{cases}$$

  with $\bar{U}$ the long-term average of both processes, $\kappa > \nu > 0$ are the mean reversion rates for both processes and $\sigma, \nu > 0$ represent the volatility of both processes.

- Why does it matter ?
  1. In the case of stablecoins, the price will revert to a value but that value does not have to be 1.
  2. It allows for considering confidence in the mean-reversion: when assets depeg they may experience long period without return to the mean

Parameters Estimation

Corollary 1

A stationary representation of the process $(S_t)_t$ is:

$$S_t = \bar{U} + \frac{\kappa}{\kappa - \eta}\nu \int_{-\infty}^{t} \left(e^{-\eta(t-s)} - e^{-\kappa(t-s)}\right) dW_s^U + \sigma \int_{-\infty}^{t} e^{-\kappa(t-s)} dW_s^S.$$

In particular, the unconditional mean and covariance function are given by the following expressions:

$$\mathbb{E}[S_t] = \bar{U},$$

$$C(t,s;\kappa,\eta,\sigma,\nu) = \mathrm{Cov}(S_t, S_s) = \frac{1}{2}\frac{\kappa^2}{\eta(\kappa^2-\eta^2)}\nu^2 e^{-\eta|t-s|} + \frac{1}{2}\left(\frac{\sigma^2}{\kappa} - \frac{\kappa}{(\kappa^2-\eta^2)}\nu^2\right) e^{-\kappa|t-s|}.$$

- Estimation of the parameters using log-likelihood

$$\begin{aligned}\mathcal{LL} &= -\frac{d}{2}\log(2\pi) - \frac{1}{2}\log\left(\det\left(C(t_i,t_j;\kappa,\eta,\sigma,\nu)\right)_{1\le i,j\le d}\right) \\ &\quad -\frac{1}{2}(\mathcal{S} - \bar{U}\mathbf{1})'\left(C(t_i,t_j;\kappa,\eta,\sigma,\nu)\right)^{-1}_{1\le i,j\le d}(\mathcal{S} - \bar{U}\mathbf{1}),\end{aligned}$$

Filtering

- In the multi-level Ornstein-Uhlenbeck, the price is observed while the underlying process has to be filtered from actual data
- We can consider instead

$$(\widehat{U}_t)_t = \left(\mathbb{E}[U_t|\mathcal{F}_t^S]\right)_t,$$

  where $\mathbb{F}^S = \left(\mathcal{F}_t^S\right)_t$ represents the natural filtration associated with $(S_t)_t$
- Classical results in stochastic filtering provide ways to estimate this $(\widehat{U}_t)_t$ based on known values of S at time t.

3. Optimal Control Model

The model

- We consider a pool of 2 assets.
- $\left(q_t^0\right)_t$ and $\left(q_t^1\right)_t$: reserves in currency 0 and currency 1 in the pool
- We assume the exogenous exchange rate, S_t, to follow a multi-level Ornstein-Uhlenbeck

$$\begin{cases} dS_t = -\kappa\left(S_t - U_t\right)\, dt + \sigma\, dW_t^S, \\ dU_t = -\eta\left(U_t - \bar{U}\right)\, dt + \nu\, dW_t^U, \end{cases}$$

- Transaction sizes are labelled in the accounting currency (currency 0)
  - A part corresponding to the market exchange rate
  - A part corresponding to a markup, accounted in currency 0

Example

If a client wants to sell z coins of currency 0 at time t, then z/S_t coins of currency 1 will be offered to her and $z\delta^{0,1}(t,z)$ extra coins of currency 0 will be asked as a markup

The model

Admissible markups

The markups $\left(\delta^{0,1}, \delta^{1,0}\right)$ belong to

$$\mathcal{A} := \left\{ \delta = \left(\delta^{0,1}, \delta^{1,0}\right) : \Omega \times [0, T] \times \mathbb{R}_+^* \mapsto \mathbb{R}^2 \middle| \delta \text{ is } \mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+^*)\text{-measurable} \right\}$$

Markups process and reserves

$$dX_t = \int_{z \in \mathbb{R}_+^*} z\delta^{0,1}(t,z) J^{0,1}(dt, dz) + \int_{z \in \mathbb{R}_+^*} z\delta^{1,0}(t,z) J^{1,0}(dt, dz)$$

$$dq_t^0 = -S_t \left(\int_{z \in \mathbb{R}_+^*} z\left(J^{1,0}(dt,dz) - J^{0,1}(dt,dz)\right) \right) \text{ and } dq_t^1 = \int_{z \in \mathbb{R}_+^*} z\left(J^{1,0}(dt,dz) - J^{0,1}(dt,dz)\right)$$

The model

Transaction processes

The processes $J^{0,1}(dt, dz)$ and $J^{1,0}(dt, dz)$ have known intensity kernels given respectively by $(\nu_t^{0,1}(dz))_{t\in\mathbb{R}_+}$ and $(\nu_t^{1,0}(dz))_{t\in\mathbb{R}_+}$, verifying

$$\nu_t^{0,1}(dz) = \Lambda^{0,1}\left(z, \delta^{0,1}(t,z)\right) m(dz)$$

and

$$\nu_t^{1,0}(dz) = \Lambda^{1,0}\left(z, \delta^{1,0}(t,z)\right) m(dz),$$

where m is a measure and $\Lambda^{0,1}$ and $\Lambda^{1,0}$ are called the intensity functions of the processes $J^{0,1}(dt, dz)$ and $J^{1,0}(dt, dz)$ respectively

Example

$$\Lambda^{0,1}(z,\delta) = \frac{\lambda^{0,1}(z)}{1 + e^{\alpha^{\mathbf{0,1}}(z)+\beta^{\mathbf{0,1}}(z)\delta}} \quad \text{and} \quad \Lambda^{1,0}(z,\delta) = \frac{\lambda^{1,0}(z)}{1 + e^{\alpha^{\mathbf{1,0}}(z)+\beta^{\mathbf{1,0}}(z)\delta}}$$

Comparison with Hodl - Excess reserves

We introduce the following two processes

$$\left(Y_t^0\right)_{t\in\mathbb{R}_+} = \left((q_t^0 - q_0^0)\right)_{t\in\mathbb{R}_+} \quad \text{and} \quad \left(Y_t^1\right)_{t\in\mathbb{R}_+} = \left((q_t^1 - q_0^1)\right)_{t\in\mathbb{R}_+}$$

And the mark-to-market excess PnL at time T is given by

$$\begin{aligned} & X_T + Y_T^0 + Y_T^1 S_T \\ = \quad & \int_0^T \left\{ \int_{z\in\mathbb{R}_+^*} z\delta^{0,1}(t,z) J^{0,1}(dt,dz) \right. \\ & \left. + \int_{z\in\mathbb{R}_+^*} z\delta^{1,0}(t,z) J^{1,0}(dt,dz) - Y_t^1 \kappa (S_t - \widehat{U}_t) dt + \sigma Y_t^1 d\widehat{W}_t^S \right\}. \end{aligned}$$

Optimal Strategies

- Derive the optimal quote $(\delta^{0,1}, \delta^{1,0})$
- If we consider exponential utility function, the optimal quote are given by

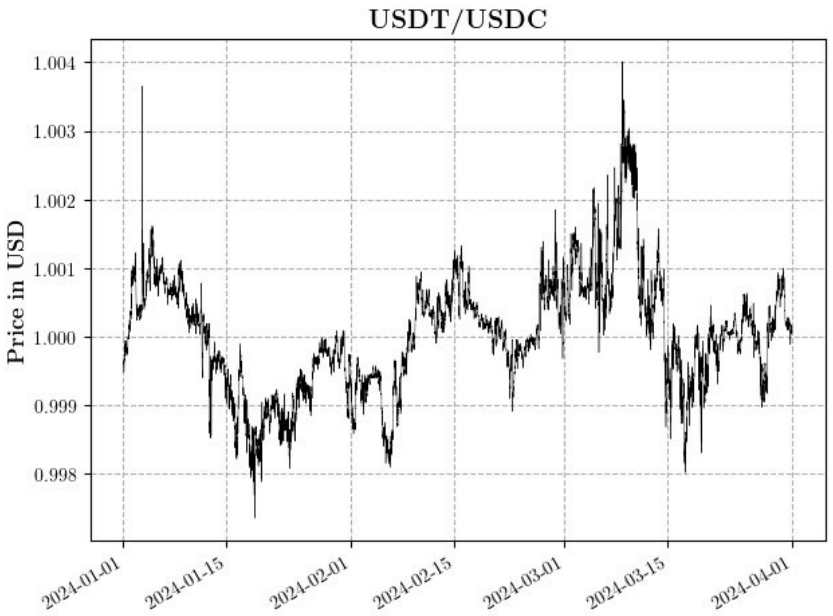
$$\sup_{\delta \in \mathcal{A}} \mathbb{E}\left[-e^{-\gamma\left(X_T + Y_T^{\mathbf{0}} + Y_T^{\mathbf{1}} S_T\right)}\right]$$

Optimal quotes

Using HJB equations and quadratic approximation for the Hamiltonians, we find explicit expressions for the optimal quotes.

4. Empirical Analysis and Simulations

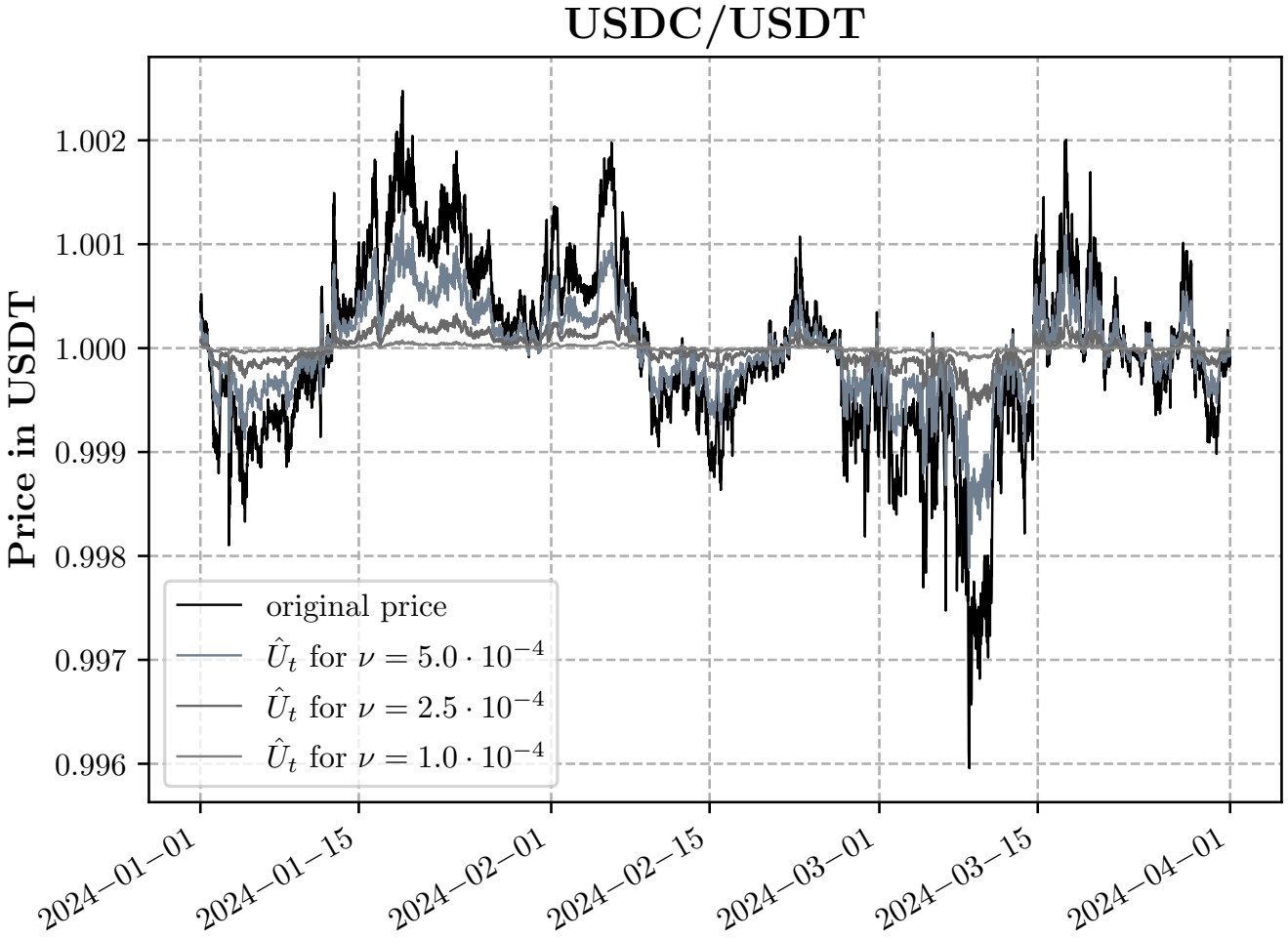
Stablecoins: estimation



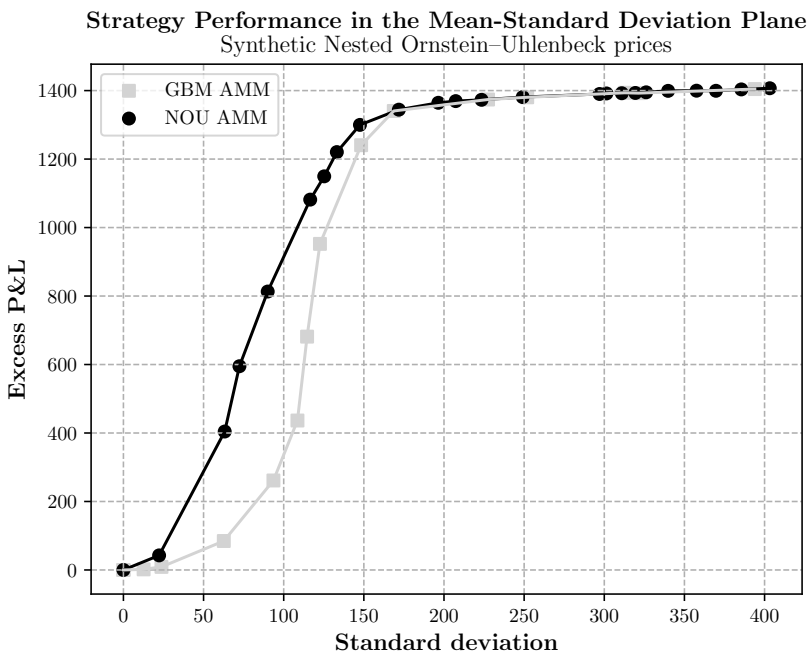
	1st-level OU		2nd-level OU		
Estimation Period	**Mean reversion**	**Volatility**	**Mean reversion**	**Volatility**	**Target**
	κ	σ	η	ν	$\bar{U}$
01-2024	4.57×10^{-2}	5.23×10^{-4}	1.6×10^{-2}	5.00×10^{-4}	1.0003
02-2024	5.46×10^{-2}	5.22×10^{-4}	5.29×10^{-2}	5.45×10^{-5}	1.0000
03-2024	4.92×10^{-2}	5.21×10^{-4}	2.96×10^{-2}	4.77×10^{-4}	0.9996

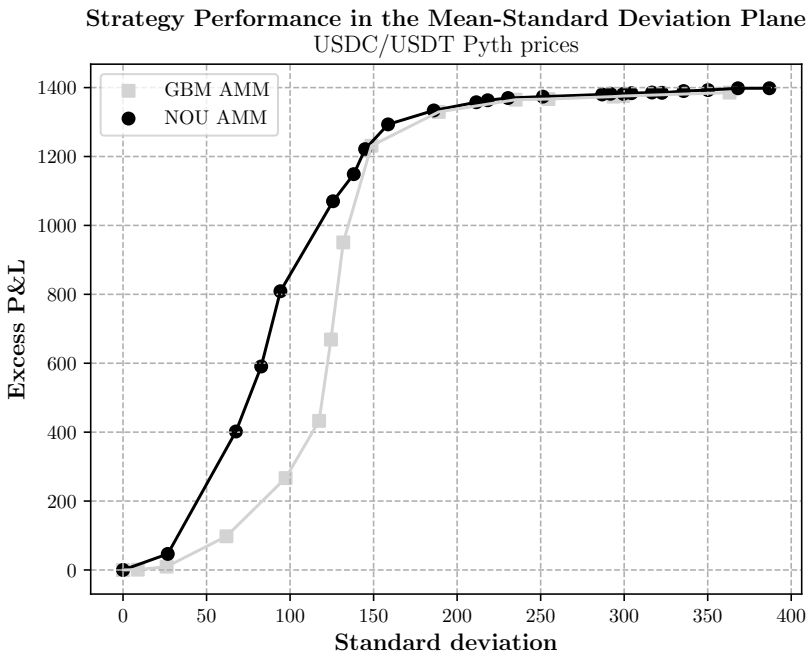
Table: Estimated parameters for the price dynamics of USDC/USDT. Units are day^{-1} for the mean reversion parameters κ and η, USDT $\cdot$ day$^{-\frac{1}{2}}$ for the volatility parameters σ and ν, and USDT for $\bar{U}$.

Stablecoins: filtering



Stablecoins: Efficient Frontier





- Simulations made using

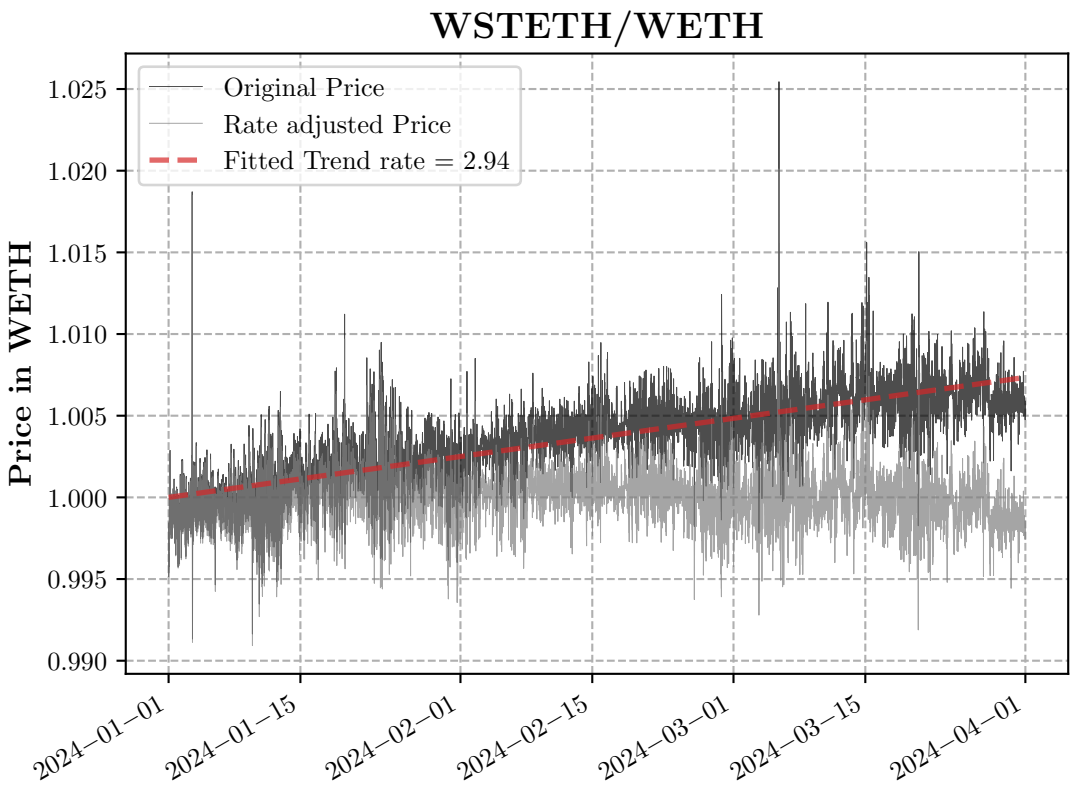
$$\lambda_{0,1} = \lambda_{1,0} = 250 \text{ day}^{-1}, \quad \alpha_{0,1} = \alpha_{1,0} = 0,$$

$$\beta_{0,1} = \beta_{1,0} = 10,000 \text{ USDT}^{-1}.$$

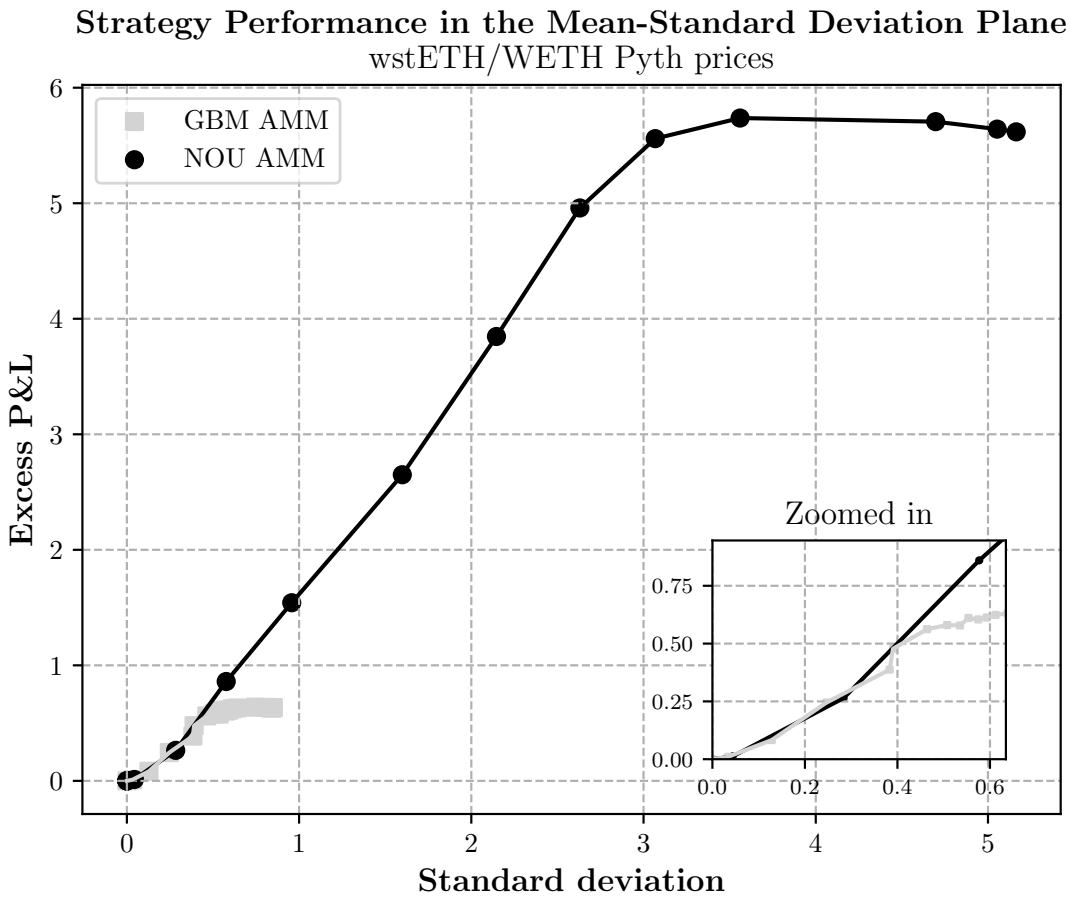
Extension to Liquid Staking Tokens (1)

- We can use the same framework for non-rebasing liquid staking tokens
- We can simply detrend the price

$$(S_t)_t = \left(\tilde{S}_t \exp\left(-\int_0^t r_s ds \right) \right)_t$$



Extension to Liquid Staking Tokens (2)



Conclusion

- Pegged assets' markets (stablecoin and LST) are some of the largest markets
- Pegged assets are better modelled by Multi-level OU than GBM
- We derive optimal bid/ask quotes under the Multi-level OU assumption
- For a pegged asset, the optimal market making strategy performs significantly better under Multi-level OU

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